Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/25/2010
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Mitigating High Z Radiation Induced Genomic Instability by Non-Protein Thiols		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	07/10/2008	**End Date:**	05/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date is 5/31/2011 per S. Krenek/JSC (7/2010). NOTE: Received extension to 5/31/2010 (from 5/31/2009) per J. Dardano/JSC (6/2009)		
Key Personnel Changes/Previous PI:	PI change to Dr. Janet Baulch, effective 7/10/2008. See Morgan, William, for fiscal year reports prior to FY2009.		
COI Name (Institution):			
Grant/Contract No.:	NNJ06HD31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	That astronauts will be exposed to the space radiation environment is unavoidable, and all radiation exposure has the potential to elicit a carcinogenic response. Cancer is characterized by gross destabilization of the genome. In vivo and in vitro exposure to radiation can lead to genomic instability. This radiation induced genomic instability shares many of the characteristics associated with genetic instabilities observed in human tumors and may well be process that transforms a normal cell to a neoplastic growth following radiation exposure. We have developed a rapid and sensitive assay for measuring induced genomic instability occurring in the progeny of irradiated cells. One way to protect space travelers from the deleterious effects of radiation exposure is with the use of radioprotective compounds. In this application we will investigate the radioprotective effect of three non-protein thiol drugs already approved for clinical use. This will be achieved by investigating three specific aims: Specific aim 1 will test the hypothesis that non-protein thiols can reduce the frequency of genomic instability induced by High Z particles in human cells. These human cells have been transfected with a sensitive and reliable reporter system to analyze delayed recombination/deletion events associated with radiation exposure in the progeny of surviving cells. Specific aim 2 will test the hypothesis that some of the delayed effects of high Z radiation result from nontargeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which the non-protein thiols exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of manganese superoxide dismutase and this reduces iron ion induced genomic instability in RKO36 cells. The goal of these studies is to determine whether drugs already approved for use in the human population can reduce genomic instability induced by radiations commonly encountered in space. If so, then these drugs can potentially be taken by astronauts to protect against deleterious effects associated with space radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The basic premise of this application is that astronauts will be exposed to space radiations because there is no way of adequately shielding the spacecraft. Thus, if delayed effects, as manifest as induced genomic instability is a consequence of high LET radiation exposure, and mediated through bystander effects and/or reactive oxygen species, then minimizing the detrimental effects of radiation exposure could be accomplished by the use of effective radical scavengers and antioxidants. While this can be accomplished using classic compounds available in the laboratory situation, e.g., DMSO, BSO etc. there are also a number of drugs used medically in the USA that are effective radical scavengers and/or antioxidants. For example, a number of thiol containing drugs, approved for clinical use e.g., amifostine (WR2721), have well established radioprotective and/or antimutagenic/anticarcinogenic properties. Should we demonstrate that any of these drugs offers statistically significant protection against the detrimental delayed effects of radiation exposure, it is possible that astronauts could take these drugs either prophylactically or immediately after a solar event and decrease the potential health hazards associated with radiation exposure.
Task Progress:	This is the year 4 report for grant NNJ06HD31G; PI change to Dr. Janet Baulch, effective 7/10/2008. See Morgan, William, for fiscal year reports prior to FY2009. The goal of this project is to evaluate the FDA approved, clinically used radio-protector amifostine for its ability to protect against the adverse effects of exposure to radiations commonly encountered in space. Additional potential radio-protective compounds are also being evaluated. The rationale for this project is that some radiation exposure is unavoidable during the course of a mission. Exposures to densely ionizing radiations can cause chromosomal instability and the subsequent genomic destabilization might well be associated with carcinogenesis. As written, the aims of the project are as follows: Specific aim 1 will test the hypothesis that NPT can reduce the frequency of genomic instability induced by High Z (HZE) particles. Specific aim 2 will test the hypothesis that some of the delayed effects of HZE radiation result from non-targeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which NPT exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of MnSOD and this reduces iron ion induced genomic instability in RKO36 cells. During the first three years of the project we evaluated the active metabolite of amifostine (WR-1065) for its ability to protect against the effects of HZE radiation exposure including genomic instability. In the publications listed below, we demonstrated that low dose WR-1065 protects against low and high LET radiation induced genomic instability and other delayed effects of irradiation (1). We also evaluated the role of MnSOD in the mechanisms of WR-1065 mediated radioprotection (2). We also demonstrated a lack of bystander response in our hands, following irradiation using electrons, x-rays, and Fe ions (3). These findings regarding bystander effects were published during year 3 of this project. The work by Dziegielewski et al. (2008) demonstrated that WR-1065 reduced radiation induced hyperrecombination using a GFP reporter system, we hypothesized that one mechanism by which WR-1065 might exert radioprotection is through suppression of homologous recombination. Using two different mammalian experimental systems, we demonstrate that WR-1065 treatment decreases the frequency of DNA damage induced homologous recombination, thereby protecting cells from the potentially negative effects of hyperrecombination (4). 1. Dziegielewski J., Baulch J.E., Goetz W., Coleman M.C., Spitz D.R., Murley J.S., Grdina D.J., Morgan W.F. (2008) WR-1065, the active metabolite of amifostine, mitigates radiation-induced delayed genomic instability. Free Radical Biology and Medicine, 12:1674-1681. 2. Murley J.S., Kataoka, Y., Baker, K.L., Diamond, A.M., Morgan, W.F., Grdina, D.J. (2007) Manganese superoxide dismuatase (SOD2)-mediated delayed radioprotection induced by the free thiol form of amifostine and tumor necrosis factor alpha. Radiation Research, 167:465-474. 3. Sowa, M., Goetz, W., Baulch, J.E., Pyles, D.N., Dziegielewski, J., Yovino, S., Snyder, A.R., de Toledo, S.M., Azzam E.I., Morgan, W.F. (2010) Lack of evidence for low-LET radiation induced bystander response in normal human fibroblasts and colon carcinoma cells. International Journal of Radiation Biology, 86:102-113. 4. Dziegielewski, J., Goetz, W., Murley, J.S., Grdina, D.J., Morgan, W.F., Baulch, J.E. (2010) WR-1065 disrupts with homologous recombination in mammalian cells. Radiation Research, 173:175-183.

Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Sowa MB, Goetz W, Baulch JE, Pyles DN, Dziegielewski J, Yovino S, Snyder AR, de Toledo SM, Azzam EI, Morgan WF. "Lack of evidence for low-LET radiation induced bystander response in normal human fibroblasts and colon carcinoma cells." International Journal of Radiation Biology. 2010 Feb;86(2):102-13. PMID: 20148696 , Feb-2010
Articles in Peer-reviewed Journals	Dziegielewski J, Goetz W, Murley JS, Grdina DJ, Morgan WF, Baulch JE. "Amifostine metabolite WR-1065 disrupts homologous recombination in mammalian cells." Radiation Research. 2010 Feb;173(2):175-83. PMID: 20095849 , Feb-2010
Articles in Peer-reviewed Journals	Dziegielewski J, Goetz W, Baulch JE. "Heavy ions, radioprotectors and genomic instability: implications for human space exploration." Radiat Environ Biophys. 2010 Aug;49(3):303-16. Epub 2009 Dec 25. PMID: 20035342 , Aug-2010